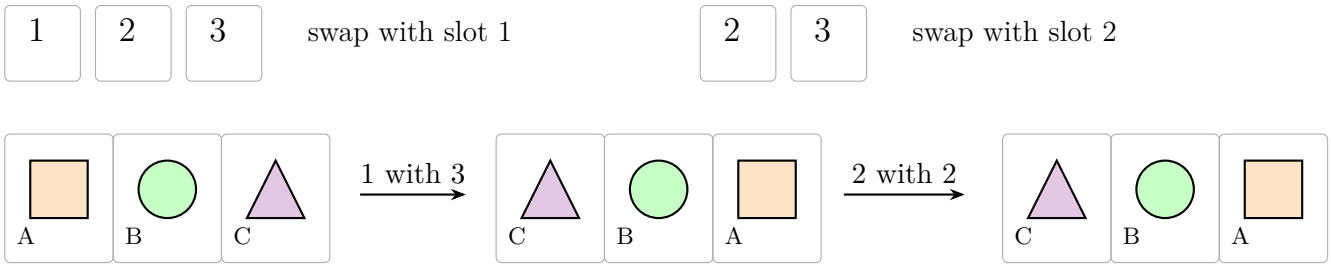
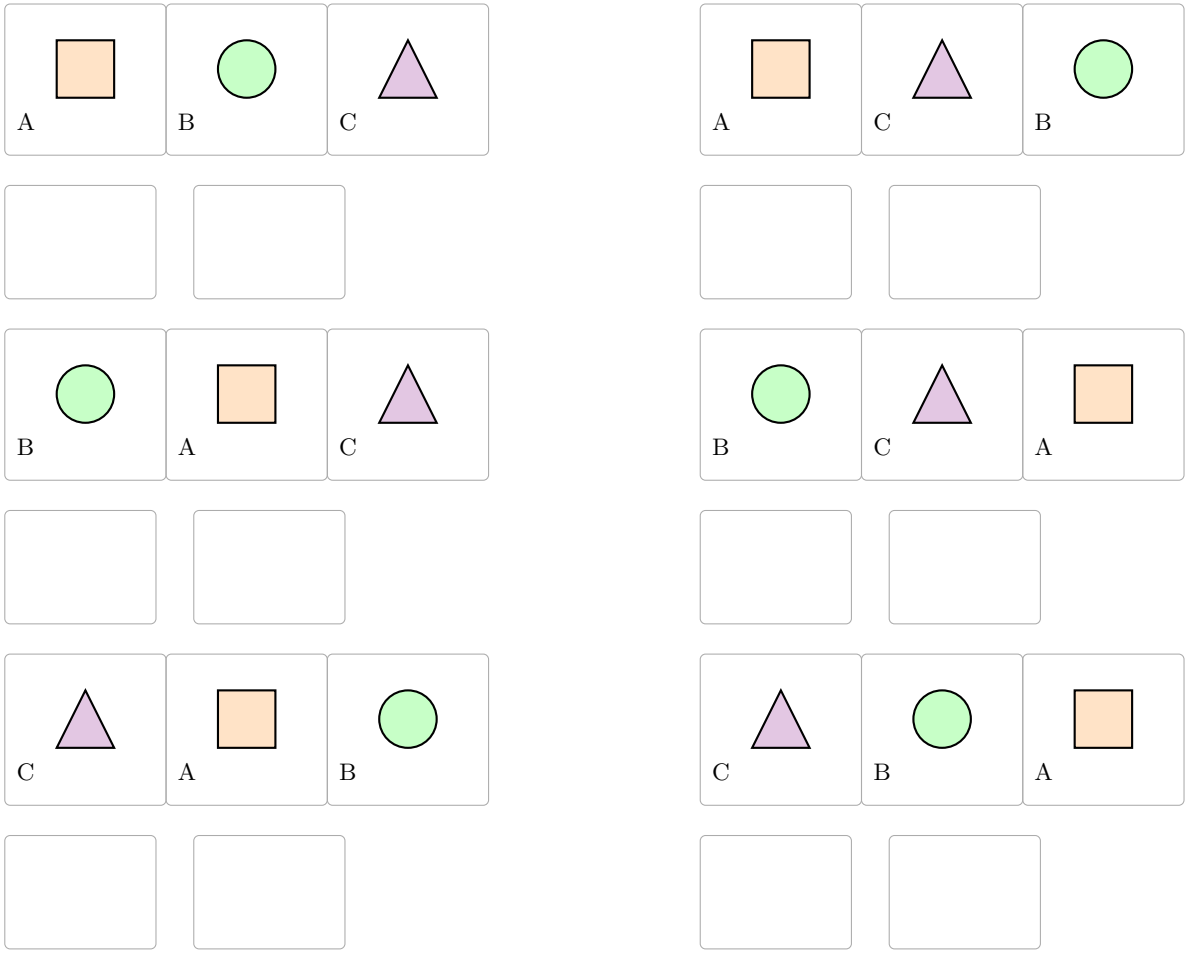
Start A B C. Mix tickets 1, 2, 3 in the cup; draw and swap its slot with slot 1. Empty the cup, then mix tickets 2, 3; draw and swap its slot with slot 2. At each draw, all tickets in the cup have equal chances. Reset both sets for every new row. A slot may swap with itself.



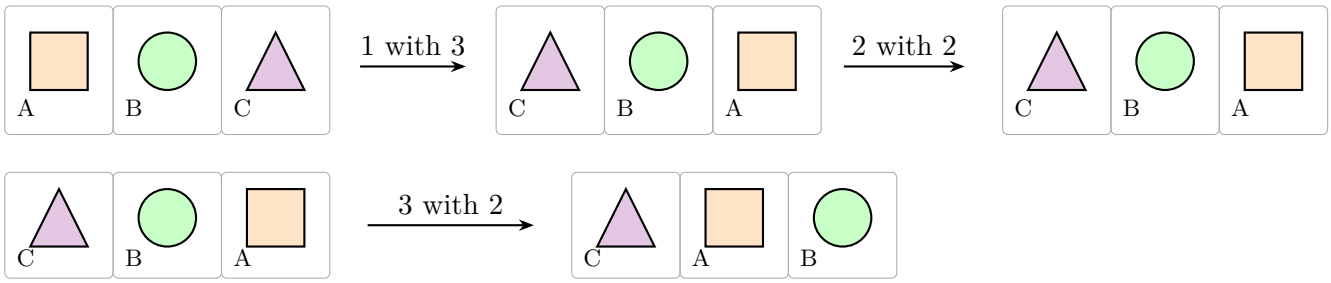
Problem 3: Start with B A C and use the same two-swap rule. Can you still make every row? Find the ticket story for each target.



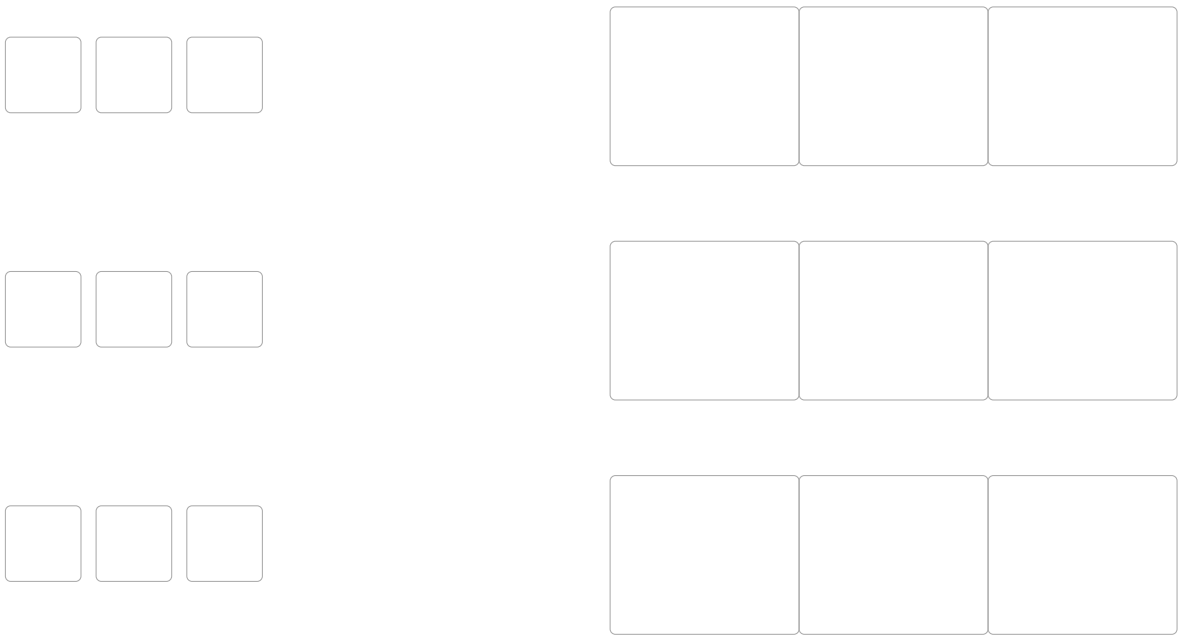
Problem 4: Shuffle six times with the two-swap rule. A friend says six shuffles must show all six rows once. Is the friend right?



Change the rule: use tickets 1, 2, 3 at every swap. Swap slot 1, then slot 2, then slot 3 with the chosen slot. Replace the ticket and mix after each draw.



Problem 5: Start with A B C each time. Find several different three-ticket stories that give the same final row.



Problem 6: There are 27 equally likely three-ticket stories. Could this rule give all six rows the same chance?